



Introduction to Data Structures and Algorithms



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Outline



1. Resources
2. What is a “Data Structure”?
3. What is an “Algorithm”?
4. Prerequisites
5. Lesson Plan
6. Grading

Resources



1. Course Materials Website:

https://github.com/noswolf/DSA_BIT/tree/DSA_24_Term2

2. Google Colaboratory or VSCode (offline) for Labs

- Interactive notebooks (.ipynb)
- <https://colab.research.google.com/>
- Login with your kmitl email address



What is a “Data Structure” ?



How do we store, organise, and retrieve data on a computer?



Abstract Data Type



- A data type where only **behavior** is defined but not implementation.
- Examples: Array, List, Map, Queue, Set, and etc.



Common vs Abstract Data Type

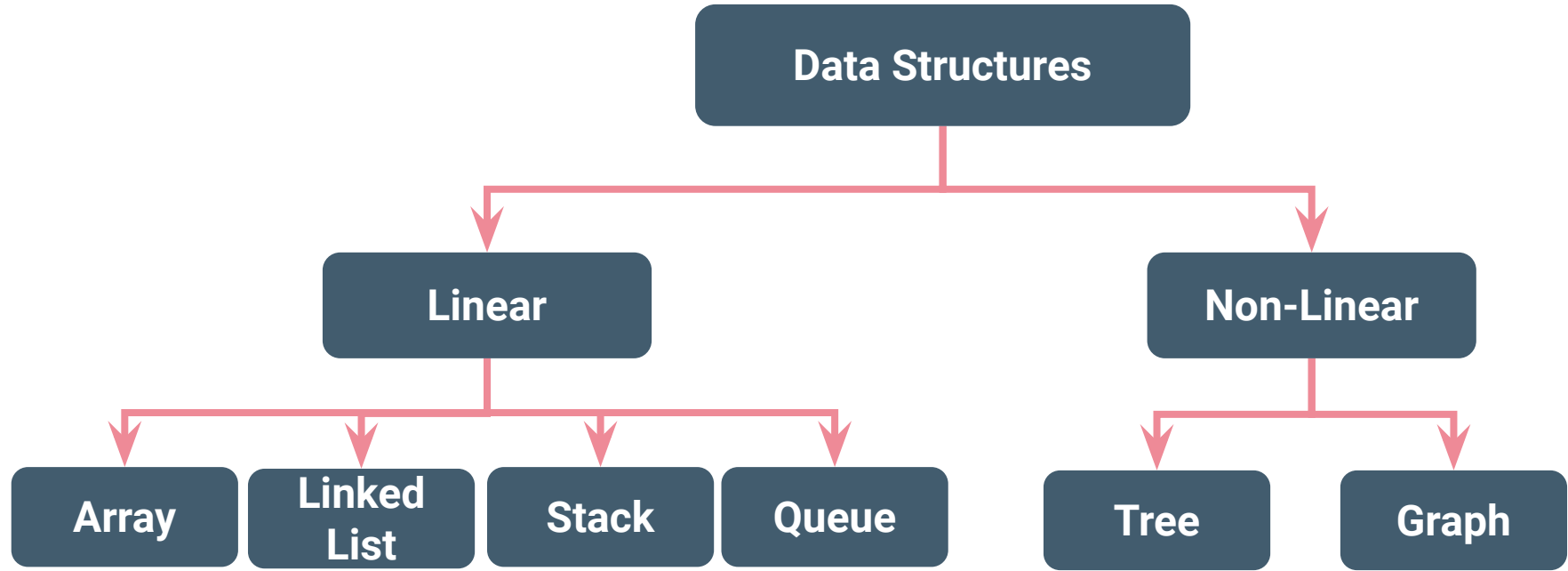
Common

- Integer
- Floating-point number
- Character
- String
- Boolean
- etc.

Abstract

- Array
- List
- Map
- Queue
- etc.

Type of Data Structure



Check out for a comprehensive list of data structures at
https://en.wikipedia.org/wiki/List_of_data_structures



What is an “Algorithm” ?





Why care about an “Algorithm” ?

How can we efficiently (in space/time) carry out some typical data processing operations?

How do we analyze and describe their performance?



Example: Sorting numbers



1. Input:
2. Sorting Algorithms
3. Output:



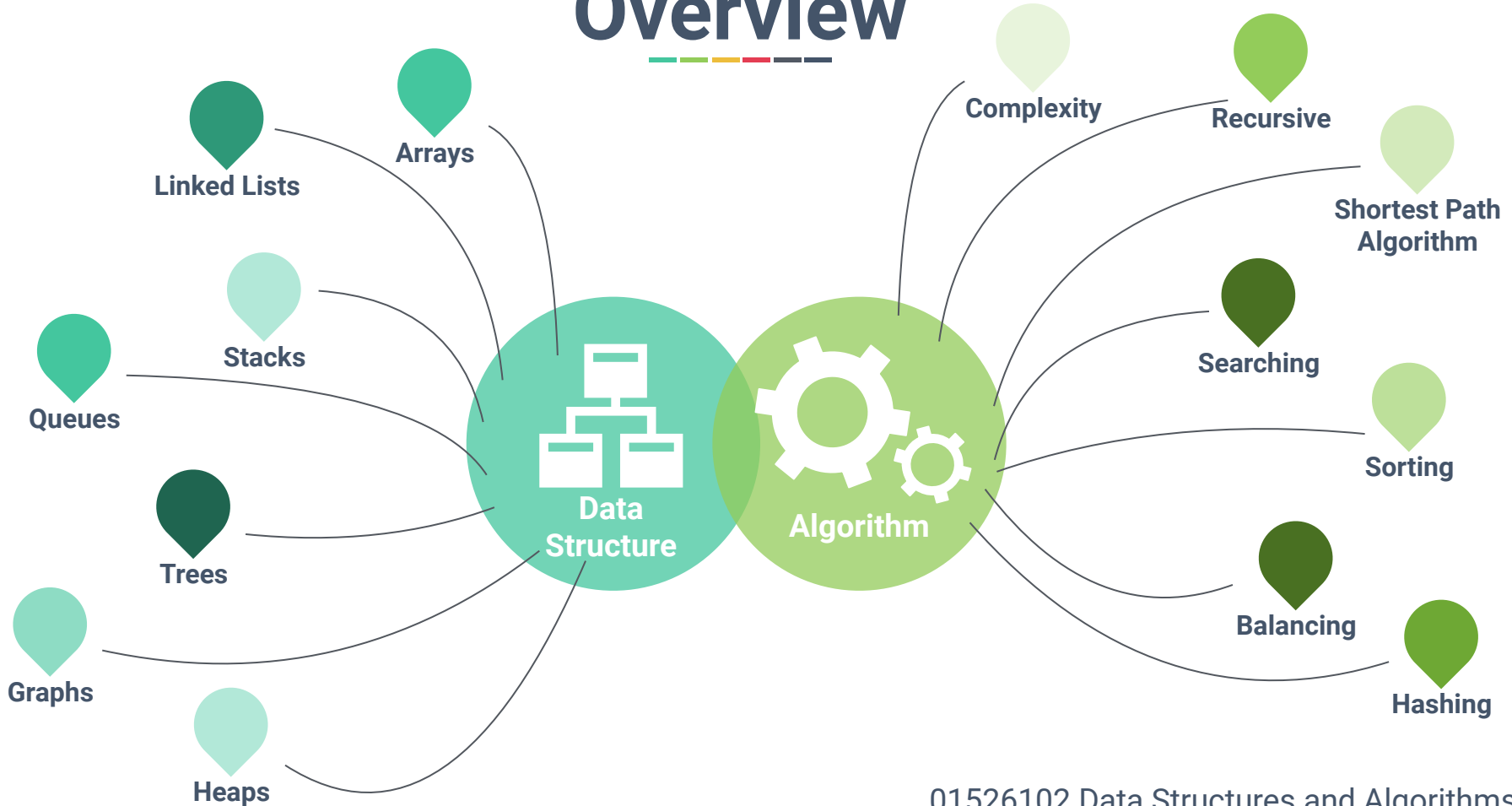
What kind of problems are solved by algorithms?



- Human Genome Project
 - identifying all genes of human beings
- Internet: Routing, searches, and security
 - **Shortest path**, search engines, encrypted communication
- E-commerce
 - Ads, recommendations, authentications
- Commercial enterprises
 - Resource allocation:
 - crew assignment on flights, package delivery route



Overview





Summary



- Data Structure
 - Way to store and organise data, allowing operations to be performed efficiently.
- Algorithm
 - Step-by-step procedure, which performs on data structure, to be followed to solve a problem/accomplish a task.



Prerequisites

- Fluent in Python Programming
- Comfortable with development processes
 - Writing a function
 - Debugging and testing a code



Lesson Plan (till Midterm)

*Public holidays are highlighted in red

Week	Topics	Labs
25/11/2024	Introduction	#1
02/12/2024	Algorithm Analysis	#2
09/12/2024	Arrays	#3
16/12/2024	Stacks	#4
23/12/2024	Queues	
30/12/2024	-	-
06/01/2025	Linked Lists	#5
13/01/2025	Linked Lists (cont.)	#6



Lesson Plan (after Midterm)

*Public holidays are highlighted in red

Week	Topics	Labs
27/01/2025	Binary Search & Hash Tables	#7
03/02/2025	Sorting	#8
10/02/2025	Recursion	
17/02/2025	Trees	#9
24/02/2025	Binary Search Trees	
03/03/2025	Graphs	#10
10/03/2025	-	-
19/03/2025	Final Exam	



Grading

Attendance	5%
Lab Assignment	25%
Midterm Exam	35%
Final Exam	35%

Attendance Score

Out of 13 times, If you attend onsite

12+ times = 5%

11 times = 4%

10 times = 2%

9 times = Unable to take a final exam

Note: Leave notice must be submitted at least 48 hrs before the lecture.

Note 2: Lab submission != Checked Attendance



Lab Assignment

Each lab required to be submitted and presented to TAs during the lab session only

- Late submission is not allowed and will not be graded
- Your TAs will be your grader, not your instructor
- If you need any help, feel free to ask me or your TAs
- MS Teams submission is required for your own benefit.
- You can ask your instructor to see a checklist of your submissions.
- Turn off AI generative features in your Colab environment



Suggestion

Working on Lab tasks

- Although this course is intense on programming, you are encouraged to work on lab with your classmates, including having intense discussion.
- If you are stuck, unsure about the task, or have no idea about the error message, feel free to ask TA and they are happy to help.



Suggestion

Generative AI age

- Even though AI features are allowed and available for you to use freely in this course, please restrain from using it as a first thing when you attempting any task.
- It can be very useful if you run out of idea/solution or need a hint from it, including help prepare you for the exam.



Things to Seriously Consider

- **Recording** my lectures in any capacity **is not allowed**.
- However, you are allowed to
 - Take photo of my handwritings on the whiteboard or
 - If you would like my VDO lectures, feel free to ask me after the lecture session.
- Alternatively, Public lectures on YouTube is a great material for you to understand the complex data structures concepts.



Reading List

Essential

Goodrich, M.T., Tamassia, R. and Goldwasser, M.H., 2013. ***Data structures and algorithms in Python***. John Wiley & Sons Ltd.

Recommended

Cormen, T.H., Leiserson, C.E., Rivest, R.L. and Stein, C., 2022. ***Introduction to algorithms***. MIT press.

Miller, B.N. and Ranum, D.L., 2011. ***Problem solving with algorithms and data structures using python***, 2nd ed. Franklin, Beedle & Associates Inc.